

# NIELS BOHR

Danish physicist  
1885–1962



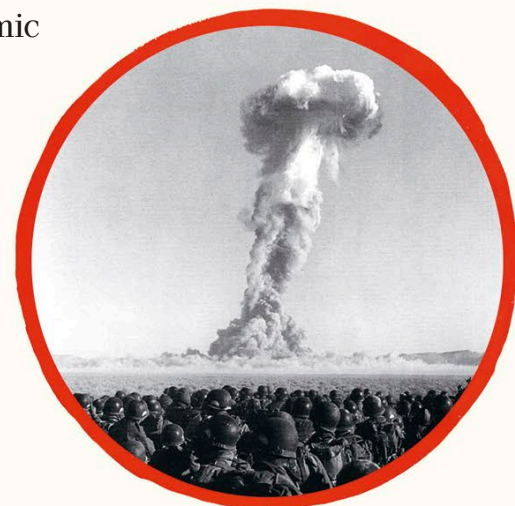
*Niels improved Ernest's model of how atoms work.*

Niels Bohr had a very different childhood from Ernest. His father was a scientist who was nominated for a Nobel Prize—twice! As a young scientist, Niels had the opportunity to work alongside Ernest at the University of Manchester in the UK. He delved deeper into Ernest's idea of the atomic structure. He focused on the electrons—the particles that orbit, or move around, the nucleus at the center of the atom. Niels taught us to think of electrons orbiting the nucleus in layers.

## Making the jump

Experiments showed that Niels's model was great at explaining puzzling things, such as why different atoms glow with different colors when heated. As atoms soak up energy, their electrons can jump up one or more layers. When the electrons return to their original places, the energy they gained is released as light.

Niels's ideas were soon being used to explain and predict characteristics of elements. With better understanding of nuclear energy, people could use it in new ways, for example, as a source of power.



*Breaking the center of atoms apart releases huge amounts of energy. Niels predicted that this could be used to make powerful bombs. The world's first nuclear bomb was detonated in 1945, during World War II.*



*After World War II, Niels worked to find nonviolent ways that nuclear physics could be used. He was one of the people who set up CERN in Europe, where scientists from all around the world study atoms and even smaller particles.*